Adham Sedik

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Education

Carleton University — Ottawa, Canada
Bachelor of Computer Science (Honours), Software Engineering Stream — *Sep 2022 – Dec 2027 (Expected)*

Technical Skills

Languages: Python, C#, TypeScript, JavaScript, Java, C++, C, Bash, HCL
Backend & Frontend: .NET 8 (ASP.NET Core), Node.js, Express, GraphQL, REST APIs; React 19, Next.js, Vite, TanStack Query
Cloud & DevOps: AWS (EKS, EC2, ECR, Lambda, IAM, VPC, S3, CloudWatch), Docker, Kubernetes, Helm, Argo CD, Kustomize, Terraform, GitHub Actions, Harness, Grafana
AI & LLMs: Claude API, Claude Agent SDK, OpenAI API, MCP servers, multi-agent orchestration, structured output schemas
Databases, Testing & Tools: PostgreSQL, DynamoDB, Redis; unit, integration and property-based testing, Playwright; Git, Jira

Experience

Solace — Ottawa, Canada
Software Developer Co-op – Platform Engineering — *Sep 2026 – Dec 2026*

- Shipped a "disabled" state for offboarded products end to end (UI, API, data model, scripts) in an internal React/Python security dashboard; disabling one stops its scans and new Jira tickets, and a nightly GitHub Actions sweep closes those already filed, **ending 30–60 minutes of hand cleanup per offboarding**.
- Made the EC2 proxies behind CI/CD builds **self-healing**: CloudWatch alarms defined in Terraform reboot an out-of-memory instance while a Lambda posts its status to Slack, turning a failure that struck a few times a month and sat unnoticed for hours into an **unattended recovery**, since proven on an actual outage.
- Provisioned an S3 artifact store in Terraform (bucket, IAM policy, versioning and lifecycle rules) and the build step that publishes a customer-facing product UI to it; now in use, it replaces fetching the latest bundle through CloudFront with **pinned releases that can be rolled back**.
- Wrote a Python reconcile check that fails, with a report, any pull request where a pipeline drifts from its Harness template, and has **blocked one such change before merge**; also added a job testing the CI scripts and refactored workflows into reusable actions.

Barracuda Networks — Ottawa, Canada
Software Engineer Co-op – BarracudaONE Platform — *Sep 2025 – Aug 2026*

- Delivered C# REST endpoints and .NET service-layer features, plus the React/TypeScript UI on top of them, over **some 60 tickets**, each covered by mock-based unit tests and gated behind a feature flag for safe rollout.
- Built a GitHub Actions workflow that reads the commit currently deployed to production and opens the hotfix branch in the right service repo, removing a manual lookup that risked branching from the wrong state; maintained in one place, it serves **18 service repos**, saving a self-timed **5–10 minutes per ticket**.
- Engineered a test-scenario generator on the Claude API and a Jira MCP server, designing the structured JSON output schema and prompts that turn story subtasks into test cases; **adopted by all 15 developers**, it cut test-note authoring time **roughly 40%**, measured before and after on real stories.
- Authored the Terraform for **production AWS infrastructure** spanning VPC, ALB, IAM, ECR and External Secrets, and set up a Grafana dashboard with threshold-based alerts on SaaS metrics, used day to day by the whole team.
- Raised governance across the service fleet: brought **FOSSA open-source license scanning** into the CI pipeline and a standard pull-request template to **every repo**, and audited three backends against agreed coding conventions.

Projects

Software factory | *Claude Code, Forgejo Actions, Bash, Playwright, k3s, Argo CD* — *2026 – Present*

- Built a self-hosted pipeline where AI agents build, review and merge my code: an orchestrator cuts each spec into small tickets, builder agents write each one test-first in its own git worktree, and a fresh reviewer agent decides whether it merges, needs me, or goes back; it has merged 100+ pull requests into lab.mediasafe.space since November 2026.
- Made unattended merges safe: the merge rules live in a script that refuses unless the latest review passed, every CI job on my own runner is green and nothing changed since the review; merges reach staging through Argo CD once the full browser suite passes, and only I promote to production.

MediaSafe | *Terraform, Kubernetes (k3s), Argo CD, Helm, Vault, GitHub Actions, Next.js, PostgreSQL* — *2026 – Present*

- Provisioned the VM, firewall, DNS and object storage with Terraform (remote state with locking, credentials injected at run time and never written to disk), and run k3s on it with **every in-cluster change made through Git**: Argo CD app-of-apps with ordered sync waves, Traefik on the Kubernetes Gateway API, cert-manager TLS and default-deny NetworkPolicies.
- Built the delivery pipeline: each merge pushes an image to GHCR and commits its tag to the GitOps repo; Argo CD runs **database migrations as a pre-rollout Job**, then a **zero-downtime rolling update**, and prod is promoted with the exact image staging ran. Secrets live in **Vault** (least-privilege Kubernetes auth) via External Secrets, snapshotted nightly with a tested restore.
- Developed the full-stack app, **live at mediasafe.space**: Next.js 16 and React over Postgres (Neon, Drizzle), Google sign-in, a shared Valkey cache and a nightly TMDB sync, with a custom ESLint rule so failed reads never render as empty.